

Serial Interface

- Considered to be one of the most basic external connections to a computer, the **serial port** has been an integral part of most computers for more than 20 years. Although many of the newer systems have done away with the serial port completely in favor of USB connections, most modems still use the serial port, as do some printers, PDAs and digital cameras.
- Essentially, serial ports provide a standard connector and protocol to let you attach devices, such as modems, to your computer.

Serial Interface

The name "serial" comes from the fact that a serial port "serializes" data. That is, it takes a byte of data and transmits the 8 bits in the byte one at a time.

The advantage is that a serial port needs only one wire to transmit the 8 bits (while a parallel port needs 8). So to send data in long distance it may be converted in serial form.

The disadvantage is that it takes 8 times longer to transmit the data than it would if there were 8 wires. Serial ports lower cable costs and make cables smaller.

Serial Interface

Serial ports, also called **communication (COM) ports**, are **bi-directional**.

Bi-directional communication allows each device to receive data as well as transmit it. Serial devices use different pins to receive and transmit data -- using the same pins would limit communication to **half-duplex**, meaning that information could only travel in one direction at a time.

Using different pins allows for **full-duplex** communication, in which information can travel in both directions at once.

Serial Interface

- Data is moved in parallel within a computer. To interface a computer with serial data lines, the data must be converted to and from serial form.
- A parallel-in-serial out shift register and a serial-in-parallel out shift register can be used to do this.

Parallel-Serial Conversion

- For Transmission, parallel data word is loaded into the shift register.
- A pulse on the clock input causes the data to be shifted.
- For an n-bit data word n clock pulses will output the word in serial form.

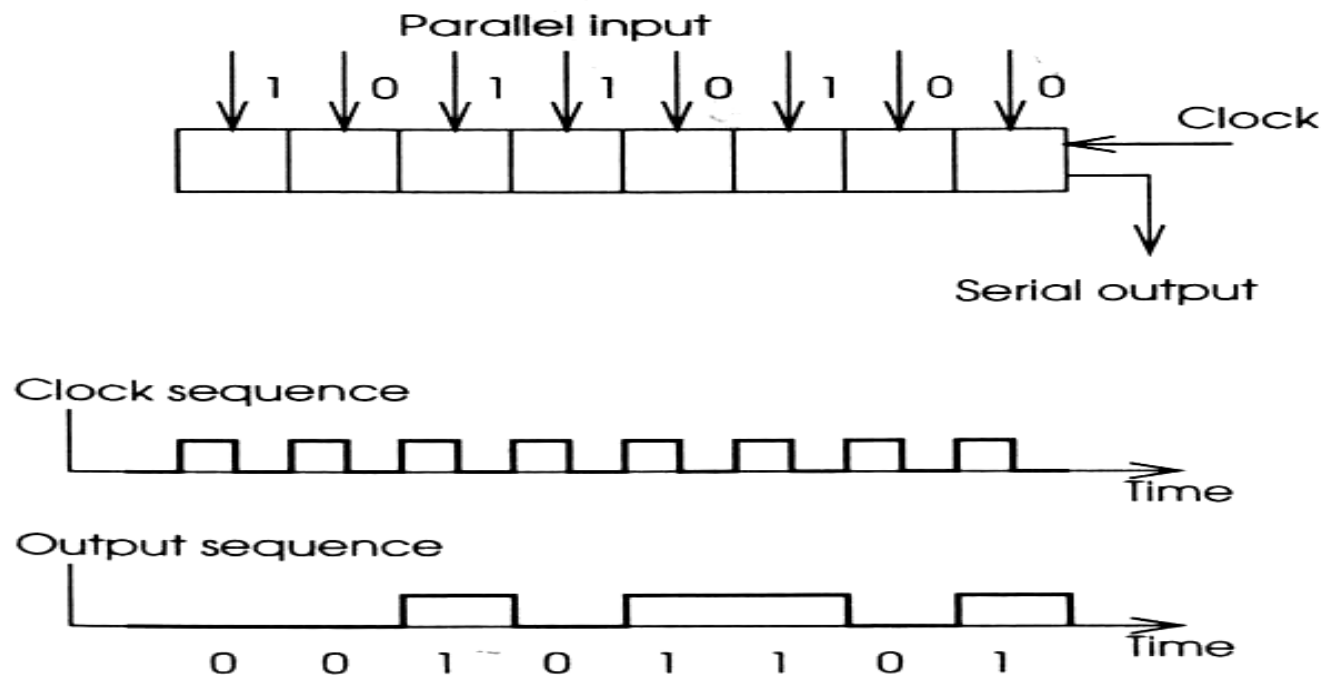


Figure 2.14 *Parallel-to-serial conversion using a shift register*

Serial-Parallel Conversion

- Reception of the serial data is performed by another shift register, in a serial-to-parallel convertor.
- A sequence of n clock pulses causes the input to propagate along the shift register until it is all available in parallel.
- The first bit to arrive is shifted all the way through the shift register and appears at the right hand end.

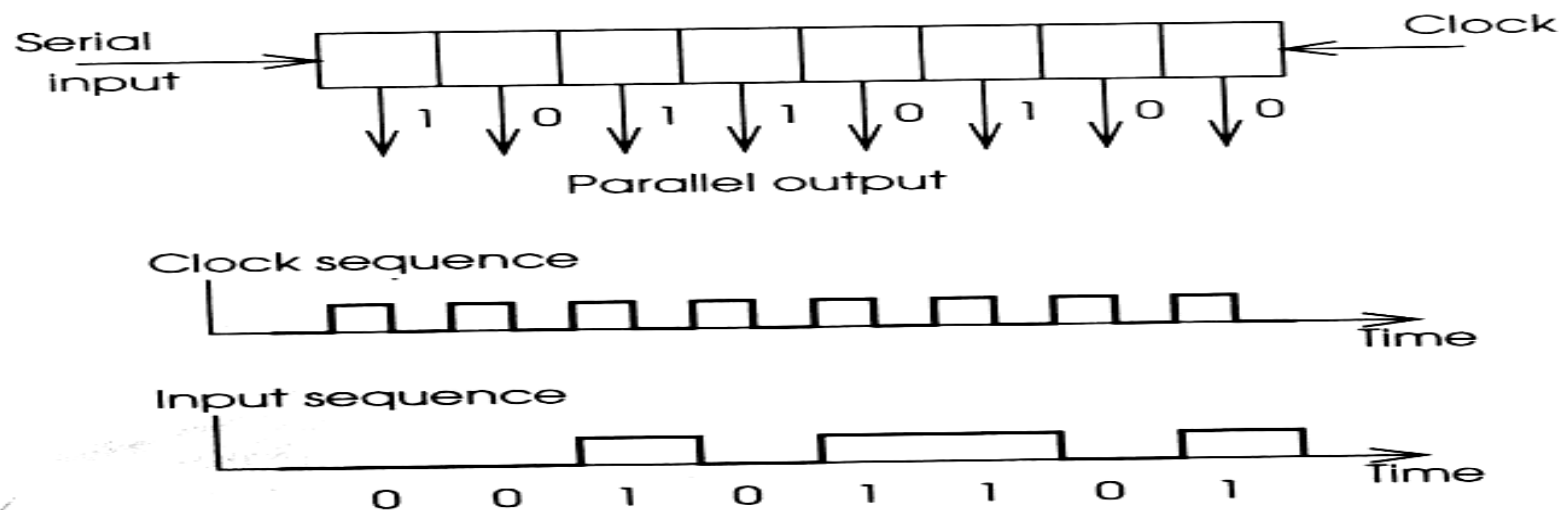


Figure 2.15 Serial-to-parallel conversion using a shift register

Serial data can be sent synchronously or asynchronously.

Asynchronous Transmission

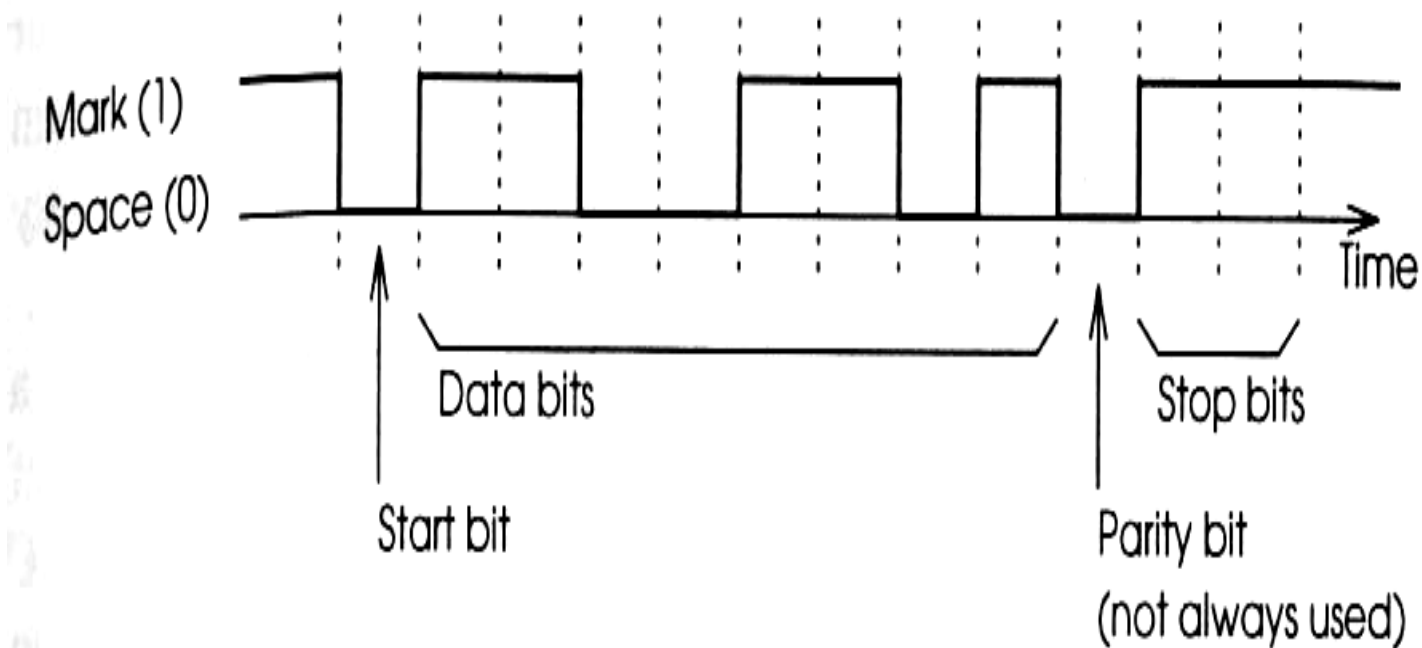


Figure 2.16 Asynchronous character format

Asynchronous Transmission

- For asynchronous transmission, characters are transmitted in the format shown in fig.
- Between characters the line is in *Idle* state which is called *Mark* or 1.
- The first bit of the character, the *Start* bit, is in the *Space* or 0.
- The 1 to 0 transition is used to start the receiver clock and indicates that a character is arriving.
- The first bit of data could not be used as a start bit because on those occasions that it was a 1, it would be indistinguishable from the idle condition.
- For the data part of the character bits are transmitted with the LSB first.
- Following the data bits there may be a parity bit to allow transmission errors to be detected.
- Finally the line returns to the idle state for 1, 1.5 or 2 stop bits which allow receiver to be ready for the next character.
- Because of this format, this is also known as stop-start transmission.

Baud Rate

- Transmission rates are sometimes expressed in bauds.
- The baud is a communication engineering unit which expresses modulation rate .
- But a signalling element does not always represent 1 bit and hence 1 baud is not necessarily 1 bit/sec.

- For example, if a signaling element can have one of the four possible voltage levels, say, 0,1,2, or 3v then each level can represent a pair of data bits, say, 00, 01, 10, 11 and 1200 baud would convey 2400 bits/sec.

Synchronous Transmission

- Data are sent continuously.
- There is no start bit to indicate where a character begins. So this must be achieved by special sequence of data known as protocol.
- One such protocol HDLC format is shown below:

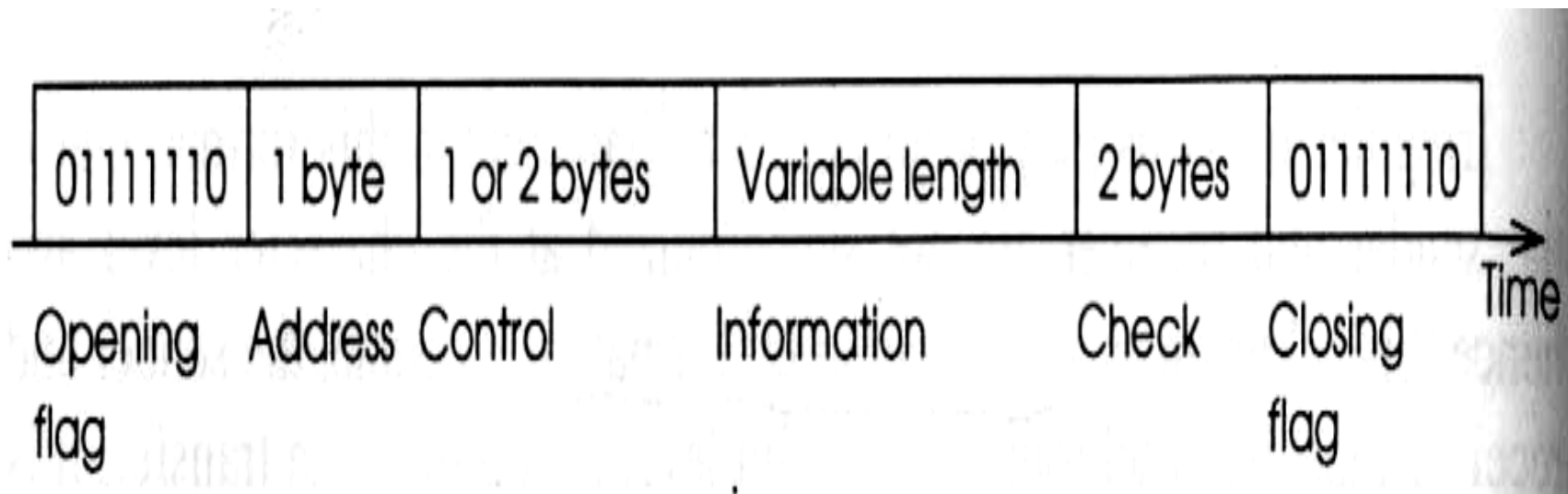


Figure 2.17 HDLC synchronous format

Synchronous Transmission

- Character synchronization is achieved by identifying a special pattern of bits (01111110) known as a Flag.
- This pattern is unique and can never occur in the data stream except as a flag.
- Since the flag contains 6 consecutive 1s, data to be transmitted is checked for 5 consecutive 1s and if found a 0 is inserted after them. This technique is called *Bit Stuffing*.

Bit Stuffing

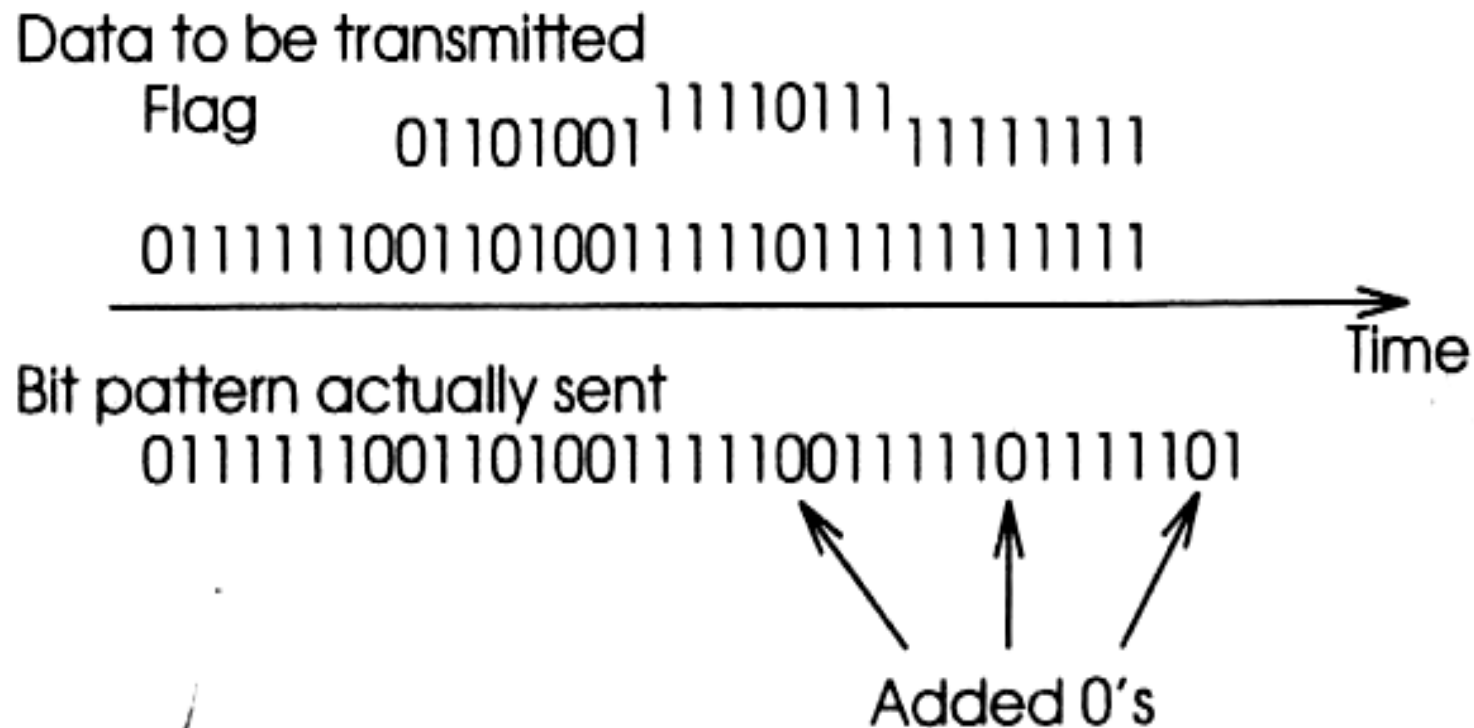


Figure 2.18 *Example of bit stuffing*

- ❑ To interface with parallel devices it is necessary to convert data to and from serial form.
- ❑ We need some handshaking circuitry to ensure that transmitter does not send faster than it can be read by the receiver.
- ❑ There are available devices which contains most of the circuitry needed for serial communication.
- ❑ Serial ports rely on a special controller chip, the **Universal Asynchronous Receiver/Transmitter (UART)**, to function properly. The UART chip takes the parallel output of the computer's system bus and transforms it into serial form for transmission through the serial port. In order to function faster, most UART chips have a built-in buffer of anywhere from 16 to 64 kilobytes. This buffer allows the chip to cache data coming in from the system bus while it is processing data going out to the serial port. But it can only do asynchronous communication.
- ❑ A device such as Intel 8251A, which can perform both asynchronous and synchronous communication, is called a Universal synchronous-asynchronous receiver-transmitter or USART.

RS232/V24 Standard

Table 2.6 RS232/V24 signal and pin allocation

Pin	V24 circuit	Abbreviation	Direction DTE:DCE	Description
1	101	PGnd	=	Protective earth
2	103	Tx	→	Transmitted data
3	104	Rx	←	Received data
4	105	RTS	→	Request to send
5	106	CTS	←	Clear to send
6	107	DSR	←	Data set ready
7	102	Gnd	=	Signal earth
8	109	RLSD	←	Received line signal detect
9				Unassigned
10				Unassigned
11	126			Unassigned
12	122	RLSD2	←	Secondary RLSD
13	121	CTS2	←	Secondary CTS
14	118	Tx2	→	Secondary Tx
15	114	TSET	←	Transmitter signal element timing
16	119	Rx2	←	Secondary Rx
17	115	RSET	←	Received signal element timing
18				Unassigned
19	120	RTS2	→	Secondary RTS
20	108	DTR	→	Data terminal ready
21		SQD	←	Signal quality detector
22	125	RI	←	Ring indicator
23	111	DSRS	←	Data signalling-rate selector
24		XTxCk	→	External transmitter clock
25				Unassigned

D-Type Connector

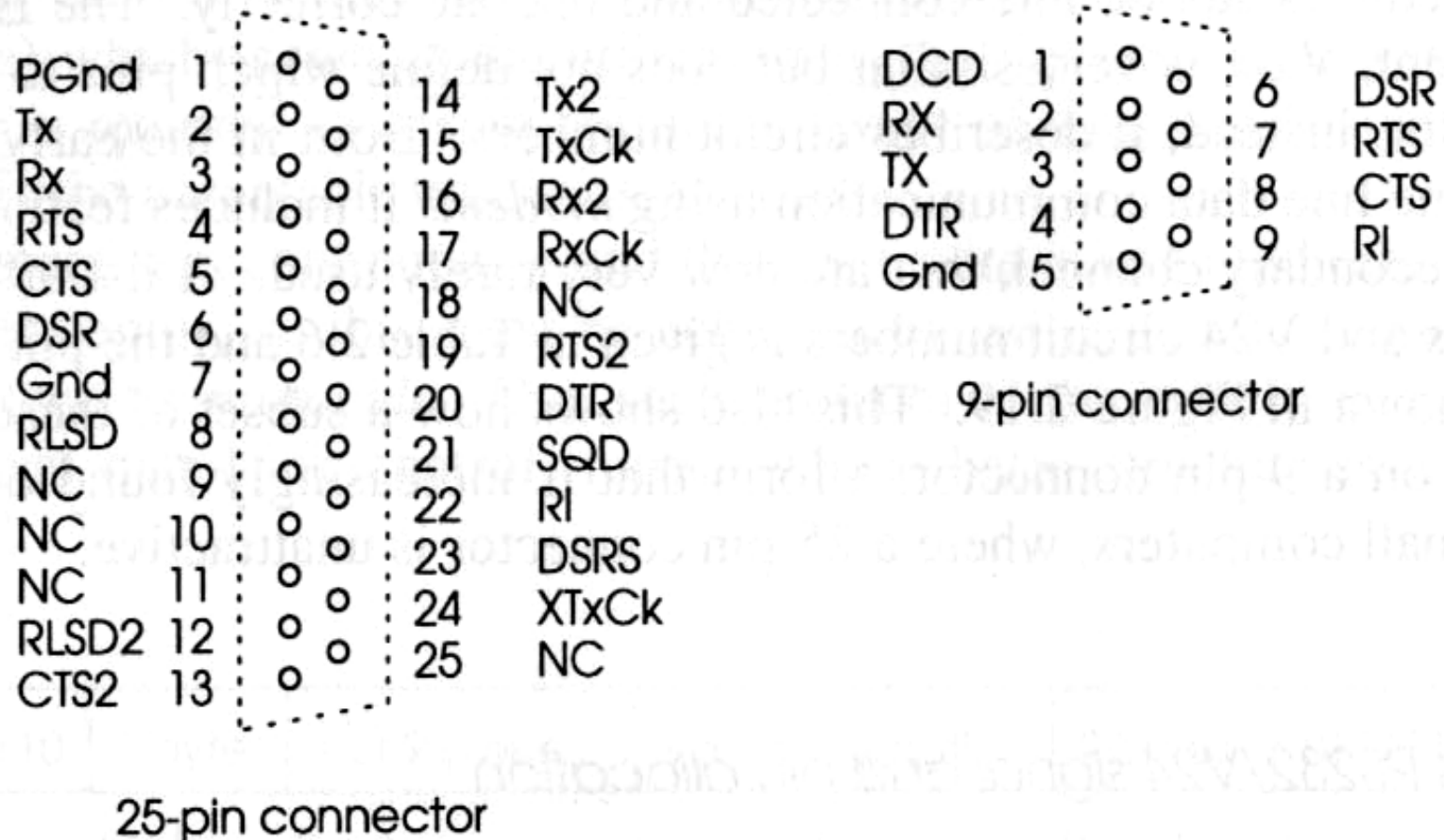
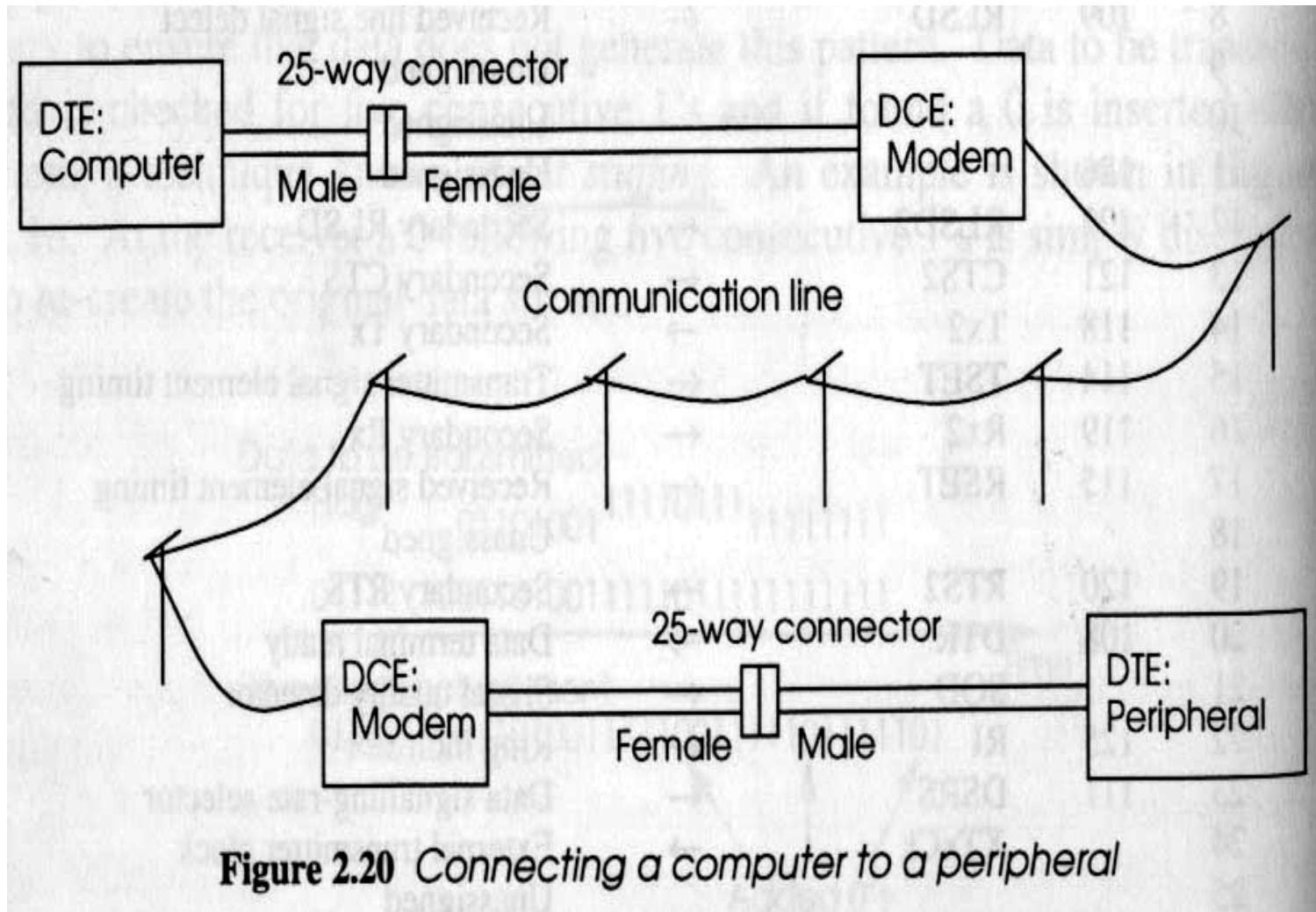


Figure 2.19 D-type connector pin assignments

Computer-Peripheral Connections



Null Modem Cable

